Question 1:

Create one database and then write SQL statements to create, in this database, all tables derived from the ERD given in the following picture with appropriate attributes, primary keys and foreign keys.

NOTE that when creating the SQL commands as request, you MUST keep the name of tables, relationship, attributes and data type of attributes as SAME as given in the given ERD.

Attributes with underline are Primary Key of each entity.

Attributes which reference to the primary key of another table must have the same name as the attributes in the primary key of the referencing table.

If a table must be created for representing a relationship, the name of the table must be the same as the name of the relationship.

When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do not use "create database", "Alter database", "use database_name" or any statements using database's name in your submission.

You must use ER style conversions for subclasses in this diagram if exists.

The table represents the multi-values attribute must have the name which is the concatenation of the table name and the attribute name. For example, if table named Student has the multi-values attribute named Phone, the name of the corresponding table must be StudentPhone.

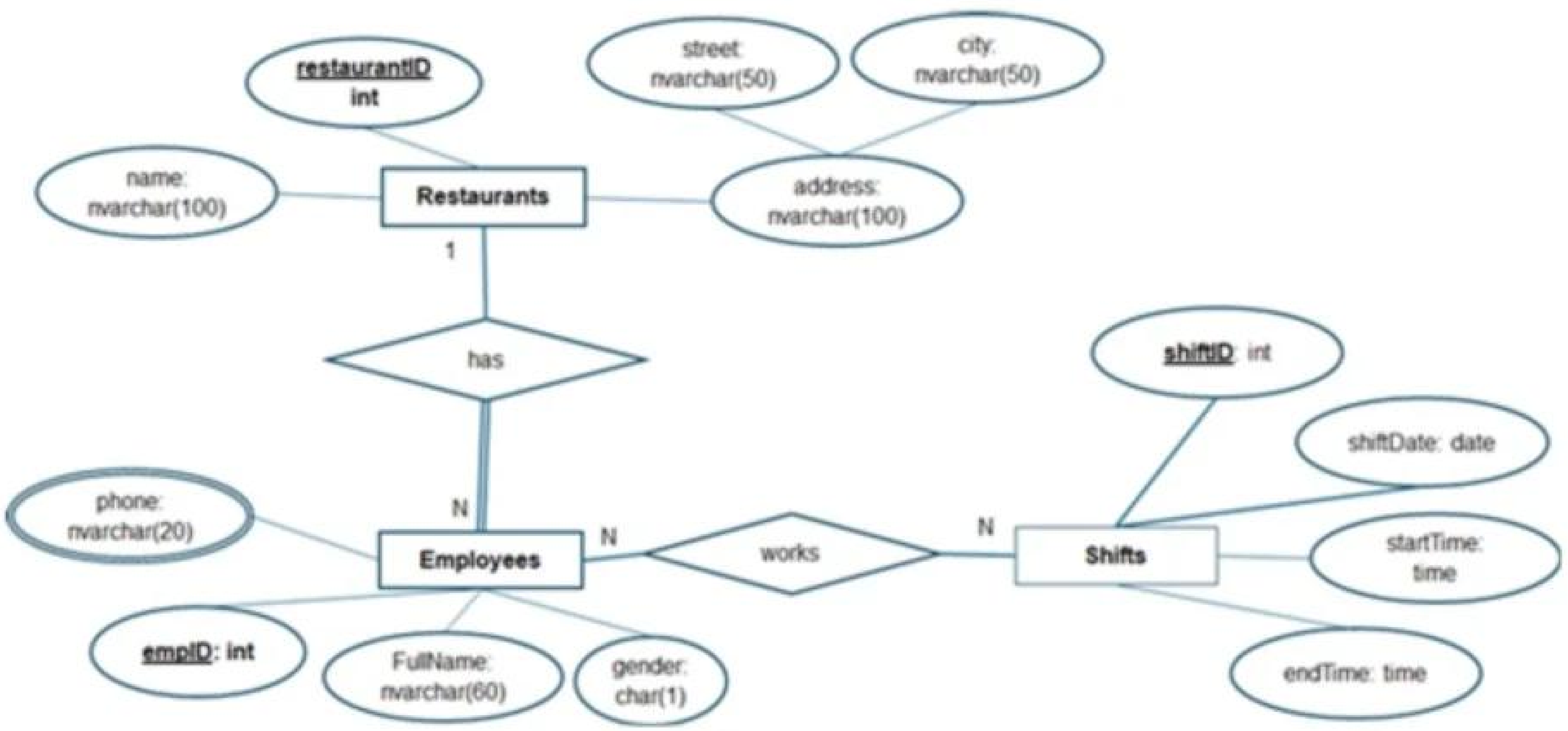
the attributes in the primary key of the referencing table.

If a table must be created for representing a relationship, the name of the table must be the same as the name of the relationship.

When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do not use “create database”, “Alter database”, “use database_name” or any statements using database’s name in your submission.

You must use ER style conversions for subclasses in this diagram if exists.

The table represents the multi-values attribute must have the name which is the concatenation of the table name and the attribute name. For example, if table named Student has the multi-values attribute named Phone, the name of the corresponding table must be StudentPhone.

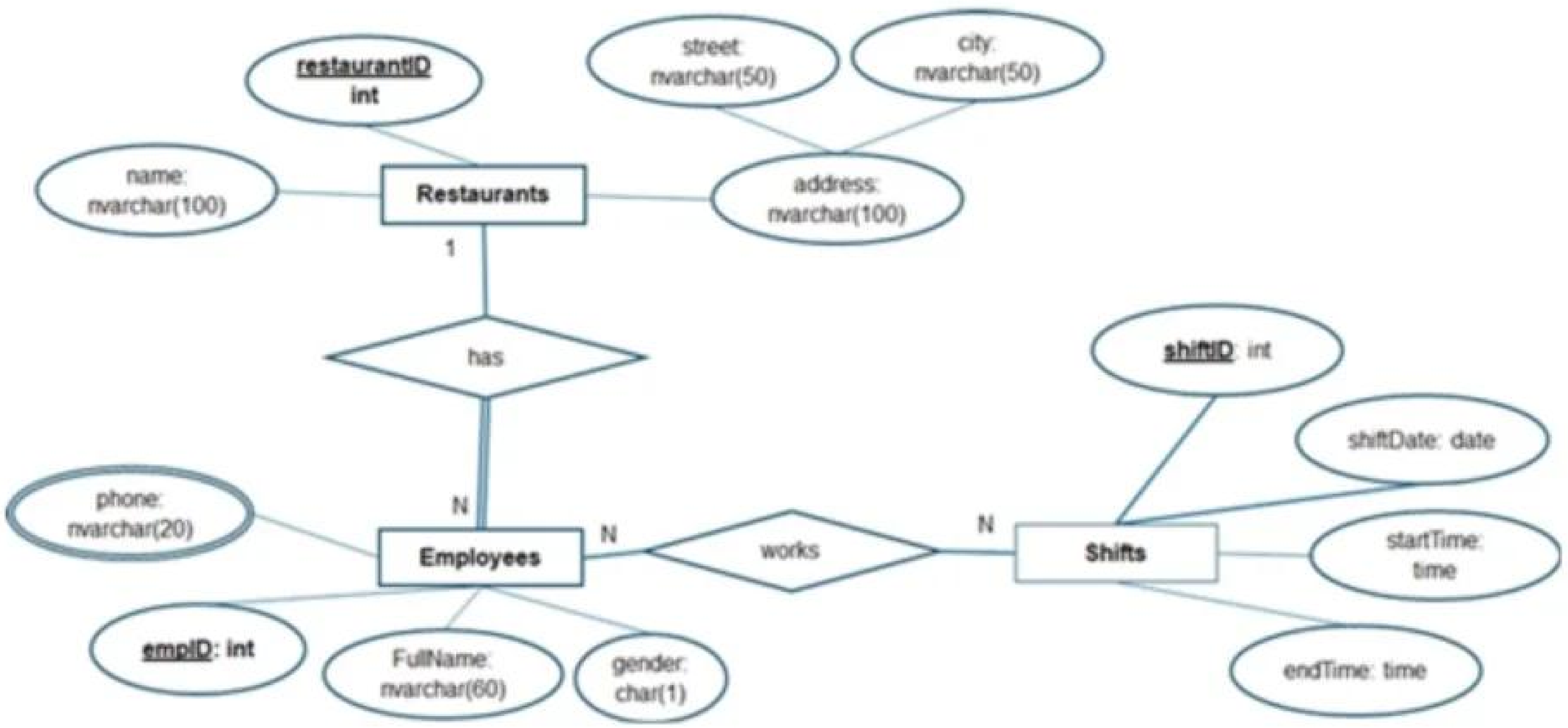


Picture 1.1

Question 2:

Write an SQL query to display all employees who are chefs or managers and were hired in 2021, as shown in the following figure:

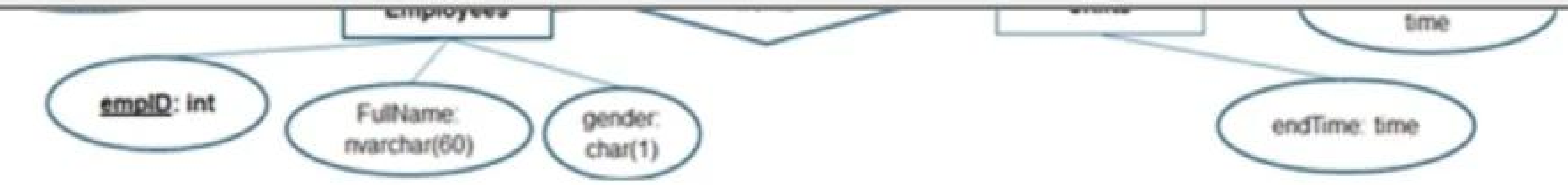
must be StudentPhone.



Picture 1.1

Question 2:

Write an SQL query to display all employees who are chefs or managers and were hired in 2021, as shown in the following figure:



Picture 1.1

Question 2:

Write an SQL query to display all employees who are chefs or managers and were hired in 2021, as shown in the following figure:

	EmployeeID	FullName	Role	HireDate	Salary
1	1	Nguyễn Văn An	Manager	2021-02-15	18000000.00
2	6	Nguyễn Thanh Tùng	Chef	2021-11-20	18500000.00
3	10	Phạm Quốc Việt	Chef	2021-08-09	19000000.00

Picture 2.1

Question 3:

Write an SQL query to list all menu items ordered by the customer named “Nguyễn Hoài Phương”, displaying the CustomerID, FullName, ItemID, and ItemName. Ensure that duplicate records are not repeated in the result:

	EmployeeID	FullName	Role	HireDate	Salary
1	1	Nguyễn Văn An	Manager	2021-02-15	18000000.00
2	6	Nguyễn Thanh Tùng	Chef	2021-11-20	18500000.00
3	10	Phạm Quốc Việt	Chef	2021-08-09	19000000.00

Picture 2.1

Question 3:

Write an SQL query to list all menu items ordered by the customer named "Nguyễn Hoài Phương", displaying the CustomerID, FullName, ItemID, and ItemName. Ensure that duplicate records are not repeated in the result:

	CustomerID	FullName	ItemID	ItemName
1	36	Nguyễn Hoài Phương	3	Bún chả Hà Nội
2	36	Nguyễn Hoài Phương	5	Cơm rang dưa bò
3	36	Nguyễn Hoài Phương	11	Nộm đu đủ bò khô
4	36	Nguyễn Hoài Phương	14	Súp ngô gà
5	36	Nguyễn Hoài Phương	15	Súp nấm hải sản
6	36	Nguyễn Hoài Phương	16	Trà đá
7	36	Nguyễn Hoài Phương	17	Nước suối
8	36	Nguyễn Hoài Phương	19	Sinh tố bơ
9	36	Nguyễn Hoài Phương	23	Nước chanh
10	36	Nguyễn Hoài Phương	26	Kem dừa

Picture 3.1

Question 4:

Write an SQL query to list all tables whose locations include "Khu A", "Khu B" or "Khu C", Write an SQL query to list all menu items ordered by the customer named "Nguyễn Hoài Phương", displaying the CustomerID, FullName, ItemID, and ItemName. Ensure that duplicate records are not repeated in the result:

	CustomerID	FullName	ItemID	ItemName
1	36	Nguyễn Hoài Phương	3	Bún chả Hà Nội
2	36	Nguyễn Hoài Phương	5	Cơm rang dưa bò
3	36	Nguyễn Hoài Phương	11	Nộm đu đủ bò khô
4	36	Nguyễn Hoài Phương	14	Súp ngô gà
5	36	Nguyễn Hoài Phương	15	Súp nấm hải sản
6	36	Nguyễn Hoài Phương	16	Trà đá
7	36	Nguyễn Hoài Phương	17	Nước suối
8	36	Nguyễn Hoài Phương	19	Sinh tố bơ
9	36	Nguyễn Hoài Phương	23	Nước chanh
10	36	Nguyễn Hoài Phương	26	Kem dừa

Picture 3.1

Question 4:

Write an SQL query to list all tables whose locations include "Khu A", "Khu B" or "Khu C", along with the customers who reserved them in September 2025. Ensure to include tables even if no customer reserved them during that month. Show the following columns: TableID, Capacity, Location, CustomerID, FullName. Sort the result in descending order of Capacity and then by Location in ascending order for tables with the same capacity.

	TableID	Capacity	Location	CustomerID	FullName
1	8	8	Khu C - Trung tâm	7	Bùi Thị Ngọc
2	4	6	Khu B - Gần quầy bar	4	Phạm Văn Cường
3	4	6	Khu B - Gần quầy bar	24	Lê Minh Hoàng
4	7	6	Khu C - Gần sân khấu	4	Phạm Văn Cường
5	3	4	Khu A - Góc phải	28	Bùi Ngọc Linh
6	2	4	Khu A - Trung tâm	46	Phạm Văn Quang

5	36	Nguyễn Hoài Phương	15	Súp nấm hải sản
6	36	Nguyễn Hoài Phương	16	Trà đá
7	36	Nguyễn Hoài Phương	17	Nước suối
8	36	Nguyễn Hoài Phương	19	Sinh tố bơ
9	36	Nguyễn Hoài Phương	23	Nước chanh
10	36	Nguyễn Hoài Phương	26	Kem dừa

Picture 3.1

Question 4:

Write an SQL query to list all tables whose locations include “Khu A”, “Khu B” or “Khu C”, along with the customers who reserved them in September 2025. Ensure to include tables even if no customer reserved them during that month. Show the following columns: TableID, Capacity, Location, CustomerID, FullName. Sort the result in descending order of Capacity and then by Location in ascending order for tables with the same capacity.

	TableID	Capacity	Location	CustomerID	FullName
1	8	8	Khu C - Trung tâm	7	Bùi Thị Ngọc
2	4	6	Khu B - Gần quầy bar	4	Phạm Văn Cường
3	4	6	Khu B - Gần quầy bar	24	Lê Minh Hoàng
4	7	6	Khu C - Gần sân khấu	4	Phạm Văn Cường
5	3	4	Khu A - Góc phải	28	Bùi Ngọc Linh
6	2	4	Khu A - Trung tâm	46	Phạm Văn Quang
7	6	4	Khu B - Trung tâm	12	Đỗ Đức Thịnh
8	1	2	Khu A - Gần cửa sổ	NULL	NULL
9	5	2	Khu B - Cạnh tường	NULL	NULL
10	9	2	Khu C - Góc trái	NULL	NULL

1	8	8	Khu C - Trung tâm	7	Bùi Thị Ngọc
2	4	6	Khu B - Gần quầy bar	4	Phạm Văn Cường
3	4	6	Khu B - Gần quầy bar	24	Lê Minh Hoàng
4	7	6	Khu C - Gần sân khấu	4	Phạm Văn Cường
5	3	4	Khu A - Góc phải	28	Bùi Ngọc Linh
6	2	4	Khu A - Trung tâm	46	Phạm Văn Quang
7	6	4	Khu B - Trung tâm	12	Đỗ Đức Thịnh
8	1	2	Khu A - Gần cửa sổ	NULL	NULL
9	5	2	Khu B - Cạnh tường	NULL	NULL
10	9	2	Khu C - Góc trái	NULL	NULL

Picture 4.1

Question 5:

Write an SQL query to list all customers whose full name starts with "Nguyễn" or "Bùi", showing for each customer: CustomerID, FullName, the number of completed orders they made in the year 2025 (column NumberOfOrders), and the total amount spent on those orders (column TotalAmount). If a customer has no completed order in 2025, still display their name with NumberOfOrders = 0 and TotalAmount = 0.

	CustomerID	FullName	NumberOfOrders	TotalAmount
1	1	Nguyễn Thị Mai	2	1902000.00
2	7	Bùi Thị Ngọc	0	0.00
3	8	Nguyễn Hoàng Nam	1	573000.00
4	14	Bùi Văn Lâm	0	0.00

Question paper (build 07.05.20.19)

1	1	Nguyễn Thị Mai	2	1902000.00
2	7	Bùi Thị Ngọc	0	0.00
3	8	Nguyễn Hoàng Nam	1	573000.00
4	14	Bùi Văn Lâm	0	0.00
5	15	Nguyễn Thị Hồng	3	1576000.00
6	21	Bùi Thị Nhung	3	2156000.00
7	22	Nguyễn Văn Tín	1	900000.00
8	28	Bùi Ngọc Linh	1	570000.00
9	29	Nguyễn Văn Khánh	1	405000.00
10	35	Bùi Thị Trang	2	1155000.00
11	36	Nguyễn Hoài Phương	0	0.00
12	42	Bùi Thanh Bình	6	1947000.00
13	43	Nguyễn Lan Hương	0	0.00
14	49	Bùi Thị Thu Hà	3	1215000.00
15	50	Nguyễn Quốc Đạt	1	800000.00

Picture 5.1

Question 6:

Write an SQL query to list all customers whose total spending on completed orders in 2025 is below the average total spending in 2025 of all customers, including those who made no orders. For each customer, display: CustomerID, FullName, TotalAmount (the total value of all completed orders in 2025 (use 0 for customers with no orders)). Ensure calculating the average total spending based on all customers, including those with 0 spending. Sort the output by TotalAmount in descending order, then by CustomerID in ascending order for customers having the same TotalAmount.

Question paper (build 07.05.20.19)

Picture 5.1

Question 6:

Write an SQL query to list all customers whose total spending on completed orders in 2025 is below the average total spending in 2025 of all customers, including those who made no orders. For each customer, display: CustomerID, FullName, TotalAmount (the total value of all completed orders in 2025 (use 0 for customers with no orders)). Ensure calculating the average total spending based on all customers, including those with 0 spending. Sort the output by TotalAmount in descending order, then by CustomerID in ascending order for customers having the same TotalAmount.

	CustomerID	FullName	TotalAmount
1	46	Phạm Văn Quang	1315000.00
2	45	Lê Thị Tâm	1310000.00

	CustomerID	FullName	TotalAmount
1	46	Phạm Văn Quang	1315000.00
2	45	Lê Thị Tâm	1310000.00
3	13	Võ Thị Hoa	1275000.00
4	49	Bùi Thị Thu Hà	1215000.00
5	4	Phạm Văn Cường	1185000.00
6	35	Bùi Thị Trang	1155000.00
7	6	Võ Minh Tuấn	1053000.00
8	17	Lê Thị Diễm	920000.00
9	32	Phạm Ngọc Bảo	910000.00
10	22	Nguyễn Văn Tín	900000.00
11	50	Nguyễn Quốc Đạt	800000.00
12	8	Nguyễn Hoàng ...	573000.00
13	28	Bùi Ngọc Linh	570000.00
14	3	Lê Thị Hạnh	548000.00
15	20	Võ Quốc Huy	535000.00
16	16	Trần Quốc Anh	522000.00
17	2	Trần Văn Tâm	481000.00

Question paper (build 07.05.20.19)

8	17	Lê Thị Diễm	920000.00
9	32	Phạm Ngọc Bảo	910000.00
10	22	Nguyễn Văn Tín	900000.00
11	50	Nguyễn Quốc Đạt	800000.00
12	8	Nguyễn Hoàng ...	573000.00
13	28	Bùi Ngọc Linh	570000.00
14	3	Lê Thị Hạnh	548000.00
15	20	Võ Quốc Huy	535000.00
16	16	Trần Quốc Anh	522000.00
17	2	Trần Văn Tâm	481000.00
18	29	Nguyễn Văn Kh...	405000.00
19	11	Phạm Thị Thuỷ	392000.00
20	26	Đỗ Văn Đức	294000.00
21	7	Bùi Thị Ngọc	0.00
22	14	Bùi Văn Lâm	0.00
23	18	Phạm Văn Long	0.00
24	19	Đỗ Thị Hằng	0.00
25	24	Lê Minh Hoàng	0.00
26	36	Nguyễn Hoài P...	0.00
27	43	Nguyễn Lan Hư...	0.00

19	11	Phạm Thị Thuỷ	392000.00
20	26	Đỗ Văn Đức	294000.00
21	7	Bùi Thị Ngọc	0.00
22	14	Bùi Văn Lâm	0.00
23	18	Phạm Văn Long	0.00
24	19	Đỗ Thị Hằng	0.00
25	24	Lê Minh Hoàng	0.00
26	36	Nguyễn Hoài P...	0.00
27	43	Nguyễn Lan Hư...	0.00

Picture 6.1

Question 7:

Write an SQL query to analyze customer spending trends between 2023 and 2024 based on completed orders. For each customer who placed completed orders in both 2023 and 2024, calculate and display:

- CustomerID

- FullName

Picture 6.1

Question 7:

Write an SQL query to analyze customer spending trends between 2023 and 2024 based on completed orders. For each customer who placed completed orders in both 2023 and 2024, calculate and display:

- CustomerID

- FullName

- AvgTotalOrderValue2023 - the average total order value of the customer's completed orders in 2023, rounded to two decimal places.

- AvgTotalOrderValue2024 - the average total order value of the customer's completed orders in 2024, rounded to two decimal places.

- PercentageChange - the percentage growth in the average total order value from 2023 to 2024. Il is calculated as (AvgTotalOrderValue2024 - AvgTotalOrderValue2023)/ AvgTotalOrderValue2023*100. Round the percentage change to two decimal places, and append ' %' at the end.

Order the result in ascending order of CustomerID as shown in the following figure.

	CustomerID	FullName	AvgTotalOrderValue2023	AvgTotalOrderValue2024	PercentageChange
1	2	Trần Văn Tâm	530000.00	340000.00	-35.85 %
2	12	Đỗ Đức Thịnh	475000.00	525000.00	10.53 %
3	13	Võ Thị Hoa	1820000.00	446000.00	-75.49 %
4	17	Lê Thị Diễm	1070500.00	235000.00	-78.05 %
5	43	Nguyễn Lan Hương	1705000.00	680000.00	-60.12 %
6	45	Lê Thị Tâm	347000.00	1102000.00	217.58 %
7	48	Võ Đức Tài	525000.00	255000.00	-51.43 %

Picture 7.1

- CustomerID

- FullName

- AvgTotalOrderValue2023 - the average total order value of the customer's completed orders in 2023, rounded to two decimal places.

- AvgTotalOrderValue2024 - the average total order value of the customer's completed orders in 2024, rounded to two decimal places.

- PercentageChange - the percentage growth in the average total order value from 2023 to 2024. Il is calculated as (AvgTotalOrderValue2024 - AvgTotalOrderValue2023)/ AvgTotalOrderValue2023*100. Round the percentage change to two decimal places, and append ' %' at the end.

Order the result in ascending order of CustomerID as shown in the following figure.

	CustomerID	FullName	AvgTotalOrderValue2023	AvgTotalOrderValue2024	PercentageChange
1	2	Trần Văn Tâm	530000.00	340000.00	-35.85 %
2	12	Đỗ Đức Thịnh	475000.00	525000.00	10.53 %
3	13	Võ Thị Hoa	1820000.00	446000.00	-75.49 %
4	17	Lê Thị Diễm	1070500.00	235000.00	-78.05 %
5	43	Nguyễn Lan Hương	1705000.00	680000.00	-60.12 %
6	45	Lê Thị Tâm	347000.00	1102000.00	217.58 %
7	48	Võ Đức Tài	525000.00	255000.00	-51.43 %

Picture 7.1

Question 8:

Write a stored procedure named MonthlySalesSummary that generates a monthly sales report for a specified year. Your procedure must accept one input parameter: @year int - the year for which the report will be generated. The procedure must display all 12 months (from January to December), even if there are no orders in a given month. For each month, calculate and display:

- Month - in the format Month/Year

- NumberOfOrders - the total count of completed orders in that month

- TotalRevenue - the total revenue (sum of Quantity * Price) from all completed orders in that month

5	43	Nguyễn Lan Hương	1705000.00	680000.00	-60.12 %
6	45	Lê Thị Tâm	347000.00	1102000.00	217.58 %
7	48	Võ Đức Tài	525000.00	255000.00	-51.43 %

Picture 7.1

Question 8:

Write a stored procedure named MonthlySalesSummary that generates a monthly sales report for a specified year. Your procedure must accept one input parameter: @year int - the year for which the report will be generated. The procedure must display all 12 months (from January to December), even if there are no orders in a given month. For each month, calculate and display:

- Month - in the format Month/Year

- NumberOfOrders - the total count of completed orders in that month

- TotalRevenue - the total revenue (sum of Quantity * Price) from all completed orders in that month

For example, when we execute the MonthlySalesSummary procedure by using the following query, the results must be shown as in the following figure:

execute MonthlySalesSummary 2023

	Month	NumberOfOrders	TotalRevenue

- NumberOfOrders - the total count of completed orders in that month

- TotalRevenue - the total revenue (sum of Quantity * Price) from all completed orders in that month

For example, when we execute the MonthlySalesSummary procedure by using the following query, the results must be shown as in the following figure:

execute MonthlySalesSummary 2023

	Month	NumberOfOrders	TotalRevenue
1	1/2023	1	475000.00
2	2/2023	0	0.00
3	3/2023	4	3315000.00
4	4/2023	2	1390000.00
5	5/2023	1	575000.00
6	6/2023	4	2115000.00
7	7/2023	1	1105000.00

	Month	NumberOfOrders	TotalRevenue
1	1/2023	1	475000.00
2	2/2023	0	0.00
3	3/2023	4	3315000.00
4	4/2023	2	1390000.00
5	5/2023	1	575000.00
6	6/2023	4	2115000.00
7	7/2023	1	1105000.00
8	8/2023	1	1070000.00
9	9/2023	1	347000.00
10	10/2023	2	2111000.00
11	11/2023	1	525000.00
12	12/2023	0	0.00

Picture 8.1

Question 9:

Create a trigger named trg_UpdateTotalAmount on the OrderDetails table to ensure the TotalAmount column in the Orders table always stays accurate.

Whenever an OrderDetails record is inserted, updated, or deleted, the trigger should

9	9/2023	1	347000.00
10	10/2023	2	2111000.00
11	11/2023	1	525000.00
12	12/2023	0	0.00

Picture 8.1

<u>Question 9:</u>

Create a trigger named trg_UpdateTotalAmount on the OrderDetails table to ensure the TotalAmount column in the Orders table always stays accurate.

Whenever an OrderDetails record is inserted, updated, or deleted, the trigger should automatically recalculate the total amount of the affected orders as the sum of (Quantity × Price) for all its items. If all OrderDetails of an order are deleted, the total amount of this order should be set to 0.

For example, if you run the following queries on the table OrderDetails, the corresponding orders will be updated.

-- test case 1 -> TotalAmount of the orders with OrderID = 3 and OrderID = 5 will be updated

INSERT OrderDetails (OrderID, ItemID, Quantity, Price)

VALUES (3, 2, 3, 75000.00), (5,27,2,40000.00)

-- test case 2 -> TotalAmount of the order with OrderID = 6 will be updated order should be set to 0.

For example, if you run the following queries on the table OrderDetails, the corresponding orders will be updated.

-- test case 1 -> TotalAmount of the orders with OrderID = 3 and OrderID = 5 will be updated

INSERT OrderDetails (OrderID, ItemID, Quantity, Price)

VALUES (3, 2, 3, 75000.00), (5,27,2,40000.00)

-- test case 2 -> TotalAmount of the order with OrderID = 6 will be updated

delete from OrderDetails where OrderID = 6 and ItemID = 1

Question 10:

Write SQL statements to create a new reservation for a customer. The reservation should be made for CustomerID = 7, with 5 guests, on November 20, 2025, at 19:15 PM. The ReservationID should be 210.

After inserting the reservation into the Reservations table, assign the reservation to a table in the ReservationTables table. The assigned table should be the one with the smallest TableID among all tables that have a capacity greater than or equal to 5.

-- test case 2 -> TotalAmount of the order with OrderID = 6 will be updated

delete from OrderDetails where OrderID = 6 and ItemID = 1

Question 10:

Write SQL statements to create a new reservation for a customer. The reservation should be made for CustomerID = 7, with 5 guests, on November 20, 2025, at 19:15 PM. The ReservationID should be 210.

After inserting the reservation into the Reservations table, assign the reservation to a table in the ReservationTables table. The assigned table should be the one with the smallest TableID among all tables that have a capacity greater than or equal to 5.